University of Tasmania

School of Information and Technology

KIT718 – Big Data Analytics

Data analytic project- Phase 1 Report

Group 8

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Problem Statement and Background

1. Problem Statement

In the context of the United Nations' Sustainable Development Goals (SDGs), ensuring healthy lives and promoting well-being for all individuals at all ages is a critical global objective, identified as SDG Goal 3 (United Nations, n.d.). Many countries face significant challenges in achieving this goal, with various factors, such as maternal mortality, life expectancy, and access to a healthy diet, influencing their progress.

The provided datasets capture essential information about different countries' health outcomes, including mortality rates, disease incidence, and the affordability of healthy diets. By analyzing these datasets, decision-makers can assess how well different nations have progressed toward SDG Goal 3, identify areas requiring improvement, and implement strategies to ensure better health outcomes globally. With this large volume of data from multiple countries and across several years, it is crucial to leverage this information effectively to derive actionable insights for improving global health standards.

The goal is to use the provided datasets to assess the achievement levels of SDG Goal 3, understand the contributing factors influencing health outcomes, and develop a data-driven approach to assist countries in their journey toward ensuring healthy lives for all.

2. Background & Context

To effectively analyze the progress toward SDG Goal 3, several datasets have been compiled that provide insights into various health-related indicators across different countries. These datasets include country-level information on achievement levels for SDG Goal 3, as well as specific indicators such as maternal mortality, life expectancy, vaccination coverage, and the cost and affordability of healthy diets.

2.1 SDG Goal 3 Achievement Dataset:

- Context: This dataset records the achievement level of SDG Goal 3 for various countries, offering an overview of how nations are performing in relation to the health-related goals. Countries were selected based on geographic diversity and income classification, ensuring representation from different continents and economic levels, from low to high income.

- Characteristics:
  - Columns: Includes id, Country, year, and goal3-achievementlevel, representing the health achievement score for each country.
  - Size: 4368 entries.
  - Data Types: Includes integer, float, and object types.

2.2. SDG Health Indicator Dataset:

- Context: This dataset provides detailed country-level information on key health indicators, including maternal and neonatal mortality, incidence rates of diseases such as tuberculosis and HIV, and life expectancy. By analyzing these indicators, policymakers can assess the progress of individual health outcomes that directly impact a country's ability to meet SDG Goal 3.
- Characteristics:
  - Columns: Includes 13 columns, covering various health-related indicators such as sdg3_matmort (maternal mortality), sdg3_neonat (neonatal mortality), and sdg3_lifee (life expectancy).
  - Size: 4825 entries.
  - Data Types: Float, integer, and object types.

2.3. Affordability of a Healthy Diet Dataset:

- Context: Context: One critical aspect of ensuring health and well-being is access to a nutritious diet. This dataset records the cost and affordability of a healthy diet across countries. It also provides information about the percentage of the population in each country who cannot afford such a diet, offering insights into the economic barriers to achieving good health.
- Characteristics:
  - Columns: Includes 7 columns, such as Country Name, Time, Cost of a healthy diet, and Percent of population who cannot afford a healthy diet.
  - Size: 1152 entries.
  - Data Types: Object and integer types.

2.4. Historical Health Indicators Dataset:

- Context: This dataset tracks various health indicators over time, from the year 2000 to 2023, across multiple countries. This comprehensive timespan allows for trend analysis and the ability to identify patterns in health outcomes over more than two decades.
- Characteristics:
  - Columns: 28 columns, including country codes, indicator names, and values for each year.
  - Size: 1601 entries.
  - Data Types: Object types.

By combining these datasets, an in-depth analysis can be conducted to monitor progress, assess the effectiveness of interventions, and guide future efforts to improve global health outcomes. Furthermore, the insights gained from this data can inform governments, international organizations, and health practitioners, helping them to tailor strategies and allocate resources where they are needed most to meet SDG Goal 3.

3. Preliminary Questions

To gain a deeper understanding of the four datasets—Food Insecurity & Nourishment Indicators, Food Prices for Nutrition, Health-Related Indicators, and Overall Health Achievement Level—a set of vital preliminary questions was developed. These questions guide the analysis using both descriptive and predictive methods.

Descriptive Questions:

- How does the cost of healthy food correlate with health outcomes such as mortality rates and life expectancy?
- What is the relationship between health expenditure (government/private) and health outcomes such as life expectancy, neonatal mortality, and maternal mortality?
- How do food prices correlate with the overall health achievement levels in different countries?

- What other socio-economic factors, beyond food prices and food insecurity, contribute to a country's health achievement level?

Predictive Questions:

- How will the st of a healthy diet evolve over the next 5 years, given historical trends?
- What will be the future prevalence of food insecurity in the next 5 years based on current health expenditure trends?
- How will life expectancy change in the next decade based on current and projected health expenditure trends?

Data Selection

1. Country Selection

In this project, 10 countries were selected based on geographic diversity, income classification, and data availability. These countries represent a range of continents and economic levels, providing a comprehensive foundation to analyze how different countries achieve health-related goals by examining health indicators, food prices, and food insecurity across diverse regions. Moreover, after checking the data, these 10 countries have enough data for analysis.

The selected countries are presented in the table below:

Country	Continent	Income Level	Data Availability
Finland	Europe	High Income	Relatively Complete
Italy	Europe	High Income	Relatively Complete
Australia	Oceania	High Income	Relatively Complete
Canada	North America	High Income	Relatively Complete
China	Asia	Upper-middle Income	Relatively Complete
Chile	South America	Upper-middle Income	Relatively Complete
Nigeria	Africa	Low Income	Relatively Complete

Nepal	Asia	Low Income	Relatively Complete
Vietnam	Asia	Lower-middle Income	Relatively Complete
India	Asia	Lower-middle Income	Relatively Complete

Selection Criteria:

- Geographic Diversity: The countries selected come from different regions, including Asia, Europe, Africa, North America, South America, and Oceania, ensuring a wide global representation.
- Economic Classification: Countries were chosen to represent different income levels, from high-income nations such as Finland, UAE, Australia, and Canada, to upper-middle-income countries like China and Chile, as well, as well as lower-income nations such as Nigeria, Nepal, Vietnam, and India.
- Data Availability: All selected countries have relatively complete datasets across the key variables intended for analysis, including food prices, food insecurity, and health indicators.

2. Data Sources

Four datasets were provided, each contributing critical data for analyzing health-related outcomes and goals across different countries. These datasets encompass information about food prices, food insecurity, and health indicators, allowing for a comprehensive analysis of the selected countries.

- Food_Insecurity_nourishment_Indicators: This dataset includes data on food insecurity and nourishment across countries, which will be used to assess the relationship between food access and health outcomes.
- Food_Prices_for_Nutrition: This dataset provides information on the cost of healthy food and the affordability of a nutritious diet. It will help analyze the impact of food prices on overall health achievements.
- Health-related_indicators: This dataset contains health-related statistics such as maternal mortality rates, child mortality rates, life expectancy, and other critical health indicators.

- Overall_health_achievement_level: This dataset contains overall health achievement levels, as computed by international organizations, allowing for comparative analysis of countries' performance in achieving health-related goals.

3. Relevant Variables

The analysis will focus on the following key variables:

- Food Prices: The affordability and cost of healthy food, which impacts the nutrition and overall health of the population.
- Food Insecurity: The proportion of the population facing food insecurity, which has significant implications for health outcomes such as child mortality and malnutrition.
- Health Indicators: Mortality rates (including maternal and child mortality), life expectancy, and other health outcomes will be used to evaluate the health status of each country.

Data Preparation

The data preparation phase involved several crucial steps to clean, transform, and standardize the datasets for further analysis. This section outlines the steps taken to ensure that the data was consistent, accurate, and ready for exploratory and predictive analysis.

1. Loading the Datasets

Four key datasets were provided and imported into Python using Pandas:

- Food Prices for Nutrition: Information on the cost of a healthy diet.
- Food Insecurity and Nourishment Indicators: Data on food insecurity and nourishment indicators across countries.
- Health-Related Indicators: Includes data on maternal mortality, life expectancy, and other health outcomes.
- Overall Health Achievement Level: Tracks the overall health achievement level of countries over time.

Each dataset was imported as a separate DataFrame and reviewed for consistency and completeness.

2. Standardization of Column and Country Names

To ensure consistency across all datasets, several transformations were applied:

- Column Standardization: Renamed 'Country Name' to 'Country' and 'Time' to 'Year' in the food_prices and food_insecurity datasets. Similarly, 'year' was renamed to 'Year' in the health_indicators and health_achievement datasets.
- Country Name Standardization: The spelling of Viet Nam was standardized to Vietnam across all datasets to ensure uniformity during merging.

These transformations were necessary to facilitate consistent merging and analysis across the datasets.

3. Filtering for Selected Countries

For this analysis, the following 10 countries were selected based on geographic diversity and data availability: Australia, Canada, Chile, China, Finland, India, Italy, Nepal, Nigeria, and Vietnam. The datasets were filtered to include only these countries, ensuring that the focus remained on a diverse sample of nations, representing different regions and income levels.

4. Handling Missing Data

Dealing with missing data was a key aspect of the data preparation process:

- Replacement of Missing Values: Periods ('..') used to represent missing data were replaced with NaN for easier handling. Missing values were identified across several variables.
- Dropping Irrelevant Columns: The 'sdg3_hiv' column was dropped from the health_indicators dataset due to high missingness.
- Imputation: Missing values in numeric columns were filled using the median to maintain data integrity without being skewed by extreme values.

This ensured that the data was complete and ready for analysis.

5. Reshaping the Food Insecurity Data

The Food Insecurity dataset was reshaped using the melt function to organize the data into a long format. Afterward, the dataset was pivoted to create a row for each country and year, allowing for easier analysis over time.

- The column names were simplified for easier interpretation.
- Missing values in the food insecurity data were filled using the median to maintain the structure and ensure accurate results.

6. Conversion of Data Types

To ensure consistency in the time-based analysis, the 'Year' column was converted to an integer type across all datasets. This was done to avoid any errors in merging or analysis, particularly for time series modeling.

7. Merging the Datasets

Once the data was cleaned and standardized, the datasets were merged into a single DataFrame called merged_data. The merge was done using Country and Year as the primary keys. An inner join was used to ensure that only records with complete data across all datasets were included in the analysis.

8. Handling Missing Values After Merging

After merging, missing values in key columns, such as Domestic general government health expenditure and Domestic private health expenditure, were filled using the median. This step ensured that the final dataset was complete and ready for the next stages of analysis.

Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) section examines the relationships between key variables, including Food Insecurity, Food Prices, Health Expenditure, and Health Outcomes, across 10 selected countries: Australia, Canada, Chile, China, Finland, India, Italy, Nepal, Nigeria, and Vietnam. The analysis addresses specific research questions and provides visual insights into global health and nutrition trends, particularly in relation to Sustainable Development Goal 3 (SDG 3): "Ensure healthy lives and promote well-being for all" (United Nations, n.d.).

1. Descriptive Statistics

Descriptive statistics provide a comprehensive overview of the key variables, including the Cost of a Healthy Diet (CoHD), the Prevalence of Food Insecurity, and various health Indicators across the selected countries.

Viriable	Mean	Medium	Maximum
Cost of a Healthy Diet (USD)	3.14	2.33	4.54
Prevalence of Moderate or Severe Food Insecurity (%)	16.66	5	69.7
Prevalence of Undernourishment (% of population)	5.18	2.5	16.6
Life Expectancy (years)	75.53	52.31	84.53
Maternal Mortality Rate (per 100,000 live births)	122.38	2.94	1135
Government Health Expenditure (% of GDP)	4.38	0.47	9.61

Table 1: Summary Statistics of Key Variables Across Selected Countries

From the table 1, the following observations can be made:

- Cost of a Healthy Diet (CoHD): The average cost across all countries is $3.14 USD, with the lowest being $2.33 USD and the highest at $4.54 USD.
- Prevalence of Moderate or Severe Food Insecurity: On average, 16.66% of the population faces moderate or severe food insecurity, with values ranging from 5.00% to 69.70%.
- Prevalence of Undernourishment: The average undernourishment rate is 5.18%, with a minimum of 2.50% and a maximum of 16.60%.
- Life Expectancy: The average life expectancy is 75.53 years, with the lowest at 52.31 years and the highest at 84.53 years.
- Maternal Mortality Rate: The average maternal mortality rate is 122.38 deaths per 100,000 live births, with a range between 2.94 and 1135.00.
- Government Health Expenditure: On average, countries spend 4.38% of their GDP on health, with a minimum of 0.47% and a maximum of 9.61%.

1.1 Visualization of Data Distribution

The following visualizations represent the distribution of key variables across the selected countries.

Distribution of Healthy Food Costs:

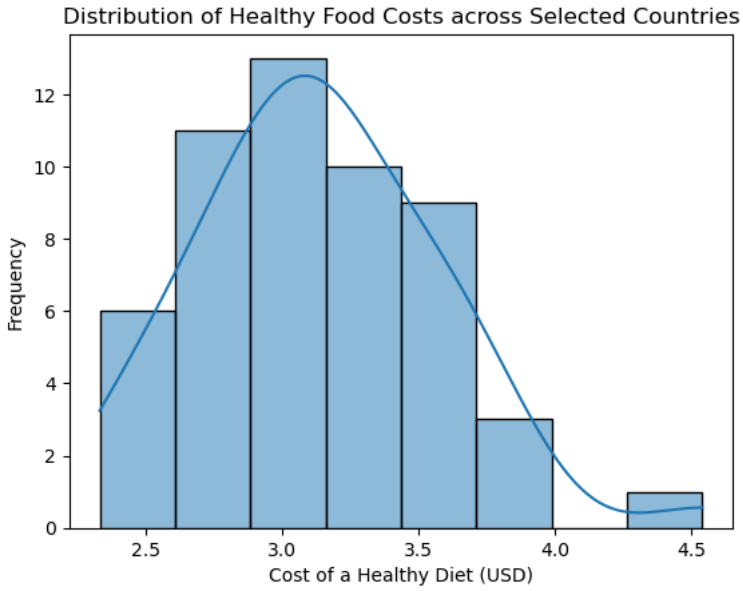

Figure 1: Distribution of Healthy Food Costs across Selected Countries.

The histogram above shows the distribution of the Cost of a Healthy Diet across the selected countries.

- The cost of a healthy diet ranges from approximately $2.33 USD to $4.54 USD, with most countries falling between $3.00 USD and $3.50 USD.
- This indicates that for the majority of the selected countries, the cost of maintaining a healthy diet is clustered in a moderate range, with a few countries having significantly higher costs.

Distribution of Prevalence of Food Insecurity:

The distribution of Food Insecurity in the selected countries is presented below.

- The prevalence of food insecurity shows significant variation, with most countries having food insecurity rates below 20%. However, a few countries experience severe food insecurity rates, reaching as high as 70%.
- This indicates substantial disparities in food access among the selected countries, with food insecurity disproportionately affecting some populations.

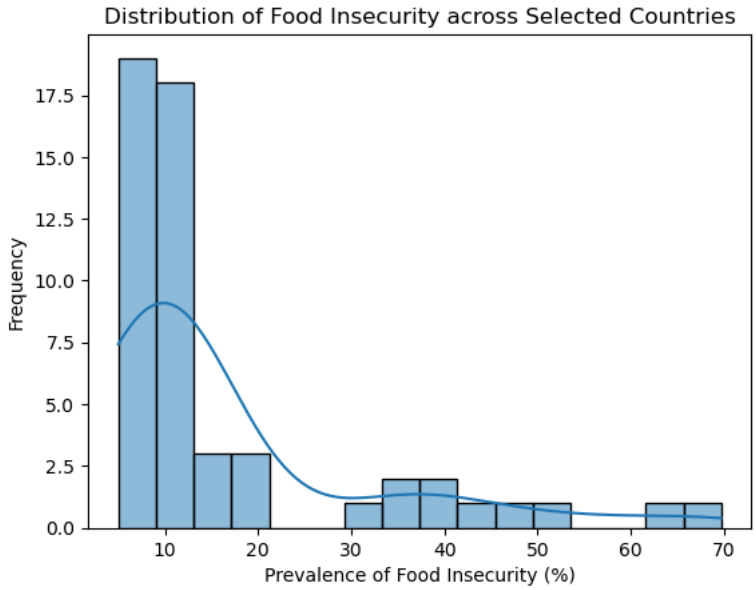

Figure 2: Distribution of Food Insecurity across Selected Countries.

Boxplot of Life Expectancy:

Figure 3: Life Expectancy across Selected Countries.

The boxplot below visualizes the Life Expectancy across the selected countries.

- Australia has the highest life expectancy, with values around 84 years, while Nigeria has the lowest, with life expectancy near 52 years.
- The boxplot also reveals the spread and distribution of life expectancy within countries. Notably, Nepal, India, and Nigeria have a wider range of life

expectancies, showing greater variability in health outcomes within these nations

Health Achievement Level Comparison

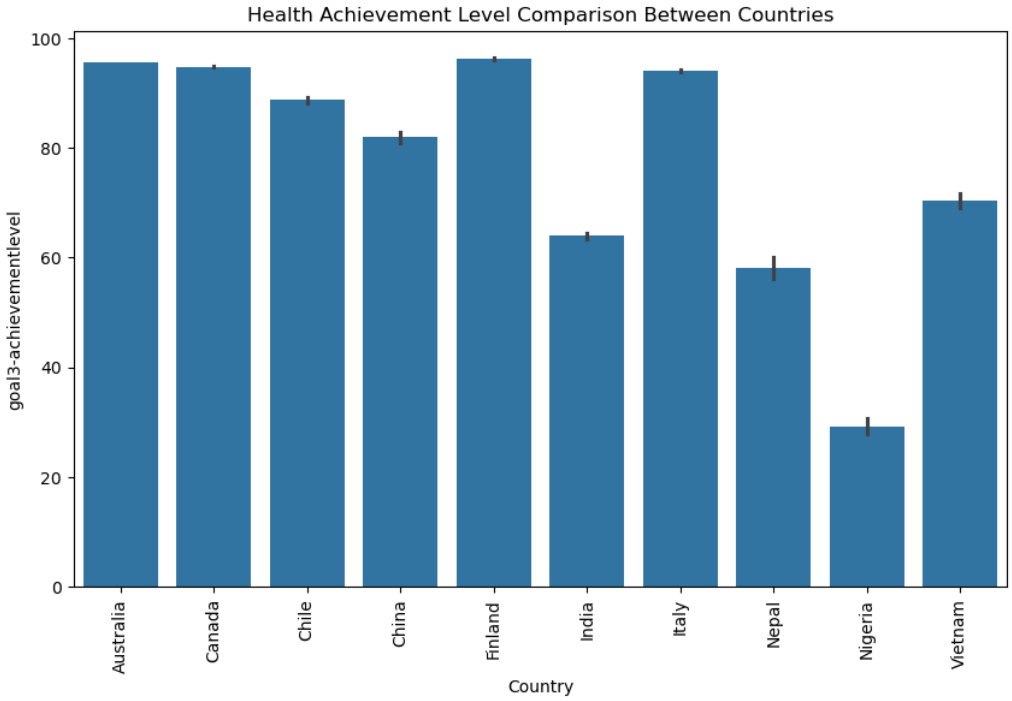

Figure 4: Health Achievement Level Comparison Across Selected Countries

The bar chart above compares the **health achievement levels** of ten selected countries: Australia, Canada, Chile, China, Finland, India, Italy, Nepal, Nigeria, and Vietnam. The health achievement level is an aggregate measure reflecting how well countries are progressing toward Sustainable Development Goal 3 (SDG 3), which focuses on ensuring healthy lives and promoting well-being for all.

- Countries such as Australia, Canada, and Finland show high health achievement levels, closely meeting SDG 3 targets.
- Nigeria and Nepal have significantly lower health achievement levels, indicating that they face substantial barriers to improving health outcomes.

1.2 Summary of Key Insights

1. Disparities in Food Access: Countries such as Nigeria and Nepal face high levels of food insecurity, while more developed countries like Canada show significantly better food access.

2. Health Outcomes: Life expectancy and maternal mortality rates vary greatly, with wealthier nations like Australia and Finland outperforming countries with lower health expenditures, such as Nigeria.
3. Health Expenditure: There is a strong link between government health expenditure and health outcomes, with countries investing more in health showing better life expectancy and lower maternal mortality rates.

2. Bivariate Analysis

The following analysis explores relationships between key economic factors (such as food prices, health expenditure, and food insecurity) and health outcomes (including life expectancy, maternal mortality, and overall health achievement levels) across selected countries. The goal is to understand how socio-economic variables influence health achievements globally.

2.1 Cost of a Healthy Diet vs. Life Expectancy

The relationship between the Cost of a Healthy Diet and Life Expectancy is analysed to determine how food prices impact longevity in various countries.

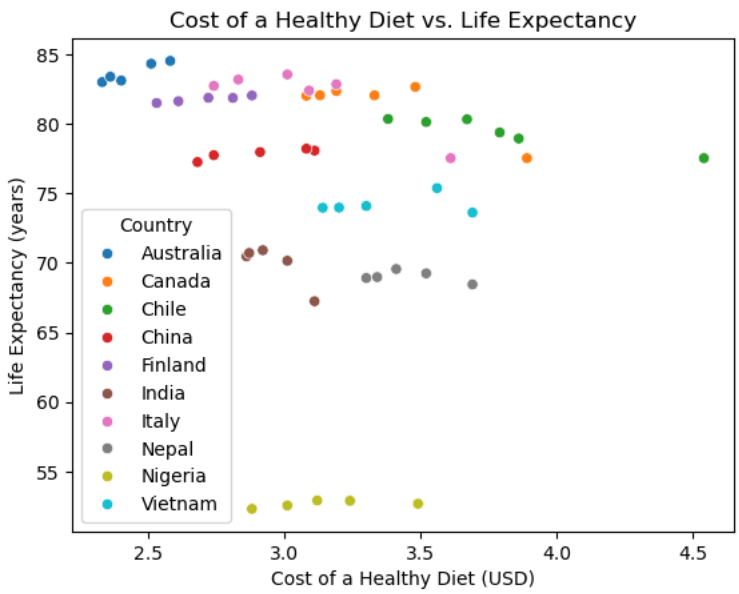

Figure 5: Relationship Between the Cost of a Healthy Diet and Life Expectancy Across Selected Countries

The scatter plot shows a weak negative correlation between food costs and life expectancy.

Countries like Australia, Finland, and Canada have high life expectancies despite higher food costs, suggesting that other factors, such as healthcare infrastructure and social policies, play a larger role in determining life expectancy.

On the other hand, Nepal and Nigeria, with lower life expectancy, do not necessarily have the lowest food costs, highlighting the complexity of factors influencing health outcomes.

2.2 Impact of Government Health Expenditure on Life Expectancy

The analysis investigates the correlation between Government Health Expenditure (% of GDP) and Life Expectancy.

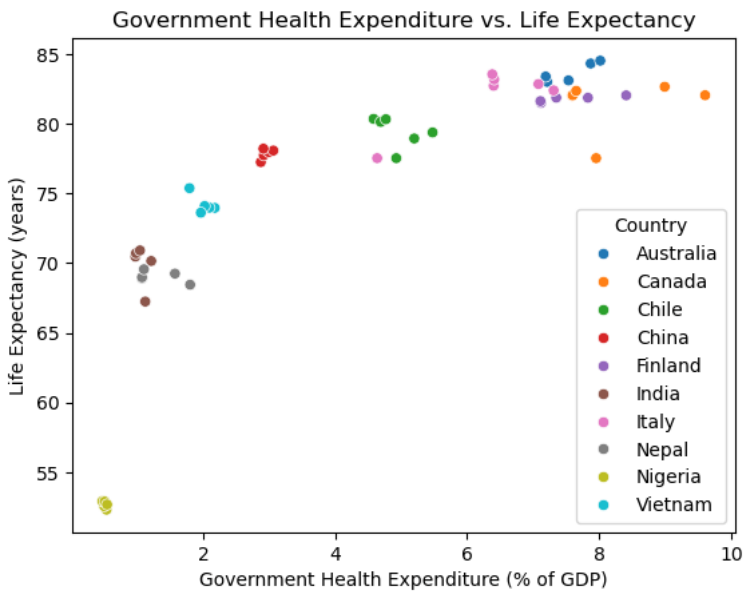

***Figure 6*: Correlation Between Government Health Expenditure and Life Expectancy Across Selected Countries**

There is a strong positive correlation between government health spending and life expectancy. Countries such as Canada, Australia, and Finland, which allocate a significant portion of GDP to health, tend to have the longest life expectancies (above 80 years). Conversely, countries like Nigeria and Nepal, with low government health spending, have the shortest life expectancies, suggesting that public investment in healthcare is crucial for improving health outcomes.

2.3 Government Health Expenditure and Maternal Mortality

The analysis display how Government Health Expenditure correlates with the Maternal Mortality Rate.

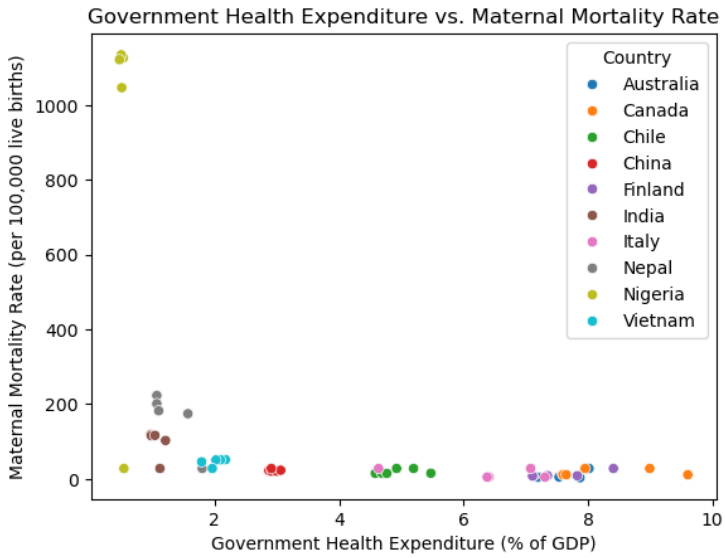

***Figure 7*: Relationship Between Government Health Expenditure and Maternal Mortality Rate Across Selected Countries**

The scatter plot demonstrates that higher government health expenditure is associated with lower maternal mortality rates. Countries like Canada, Australia, and Finland, which invest heavily in public health, have maternal mortality rates near zero. In contrast, Nigeria, with very low government health expenditure, experiences extremely high maternal mortality rates. This highlights the importance of government funding in improving maternal health outcomes, particularly in low-income countries.

2.4 Correlation Between Food Prices and Health Achievement Levels

This section explores the relationship between the Cost of a Healthy Diet and a country's Health Achievement Level

Figure 8: Relationship Between Food Prices and Health Achievement Levels Across Selected Countries

No clear correlation is observed between food prices and health achievement levels. Countries like Australia and Finland maintain high health achievement levels despite relatively high food prices, suggesting that the overall health achievements depend more on the healthcare system, public policies, and social support than just the cost of food.

2.5 Impact of Food Insecurity on Life Expectancy

The relationship between Food Insecurity and Life Expectancy is explored to understand how access to food impacts overall health.

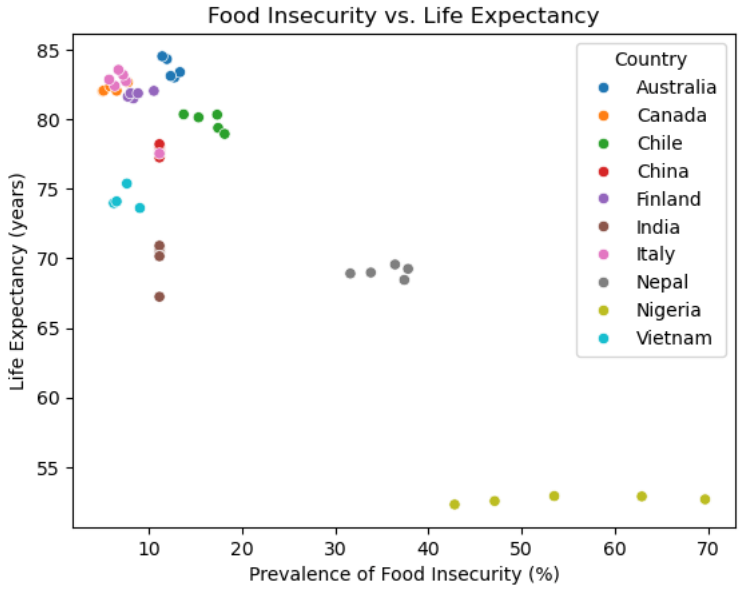

Figure 9: Impact of Food Insecurity on Life Expectancy Across Selected Countries

The scatter plot shows a strong negative correlation between food insecurity and life expectancy. Nigeria and Nepal, with high food insecurity levels, have much lower life expectancies compared to countries like Canada and Finland, where food insecurity is minimal. This highlights that access to adequate food is a critical determinant of life expectancy, particularly in low-income nations.

2.6 Food Insecurity and Maternal Mortality Rates

Finally, the correlation between Food Insecurity and the Maternal Mortality Rate is analysed.

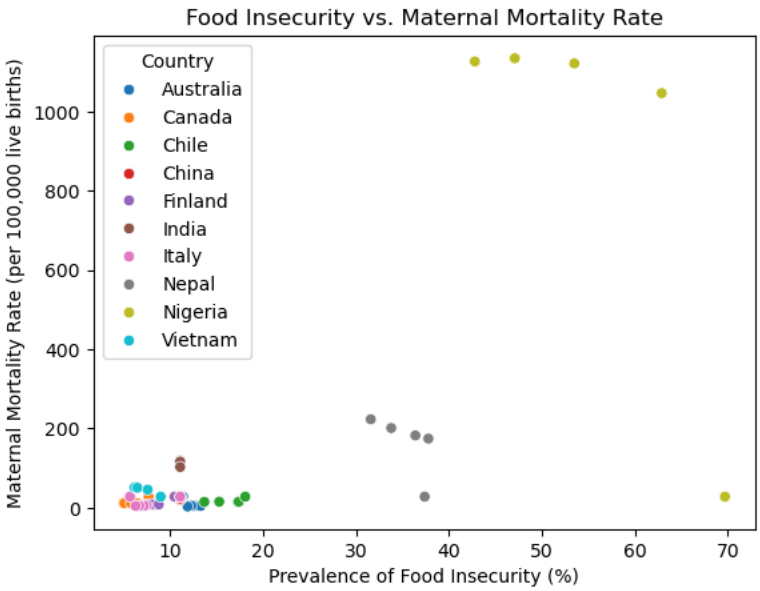

Figure 10: Correlation Between Food Insecurity and Maternal Mortality Rates Across Selected Countries

Countries with high food insecurity, such as Nigeria and Nepal, also experience the highest maternal mortality rates, indicating that food insecurity exacerbates maternal health challenges. In contrast, countries like Canada, Finland, and Australia, with minimal food insecurity, show near-zero maternal mortality rates. This suggests that ensuring food security is key to improving maternal health outcomes.

3. Predictive Analysis

The predictive analysis aims to forecast the Cost of a Healthy Diet, Food Insecurity Levels, and Life Expectancy for the next 3-5 years based on historical data trends and health expenditure patterns. The results of these predictions offer insights into the future state of food affordability, population health, and governmental impact on public well-being.

3.1 Forecasting the Cost of a Healthy Diet

Predict how the cost of a healthy diet will evolve over the next 5 years, given historical data trends.

A linear regression model was trained on historical data for the Cost of a Healthy Diet (CoHD), using Year as the independent variable. This model was used to forecast future values from 2023 to 2027.

The predicted cost of a healthy diet shows a consistent upward trend.

The cost is forecasted to rise from 3.70 USD in 2023 to 4.29 USD by 2027.

The chart below illustrates the historical data (blue), the linear trend (green), and future predictions (red).

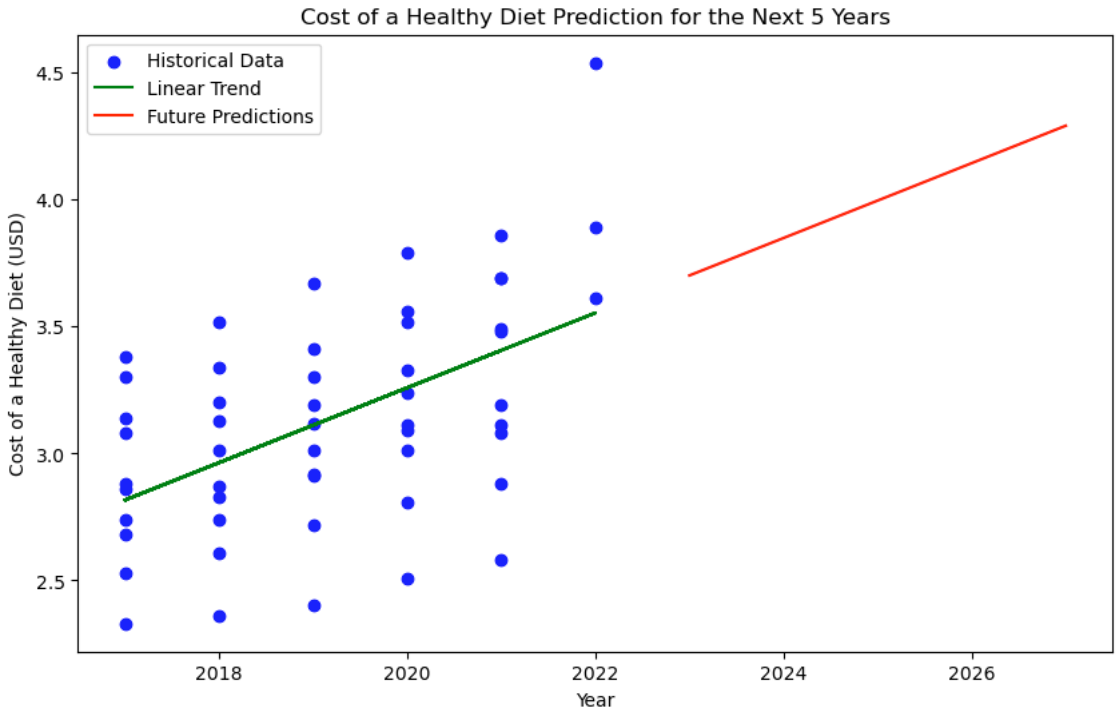

Figure 11: Cost of a Healthy Diet Prediction for the Next 5 Years

	Year	Predicted Cost of Healthy Diet
0	2023	3.701663
1	2024	3.849191
2	2025	3.996719
3	2026	4.144247
4	2027	4.291774

Table 2: Predicted Cost of a Healthy Diet (2023 - 2027)

The forecast indicates a steady increase in the cost of a healthy diet. If this trend continues, it may become more challenging for lower-income populations to afford nutritious food, which could have broader implications for public health and food security.

3.2 Forecasting Food Insecurity Based on Health Expenditure

This analysis aims to forecast the future prevalence of Food Insecurity over the next three years, based on projected trends in health expenditure. Understanding the relationship between health expenditure and food insecurity allows policymakers to

predict how increasing government investments in health might influence food security.

A linear regression model was used to examine the relationship between Current Health Expenditure (% of GDP) and the Prevalence of Moderate or Severe Food Insecurity. The model was applied to predict future food insecurity levels for projected health expenditure values in the years 2024, 2025, and 2026.

The model predicts a reduction in food insecurity as health expenditure increases. At a 7% health expenditure, food insecurity is projected to be 17.34%. As health expenditure increases to 9%, food insecurity is expected to decline further to 12.37%. The scatter plot below visualizes the historical data (blue) and future predictions (red) for food insecurity levels based on projected health expenditure.

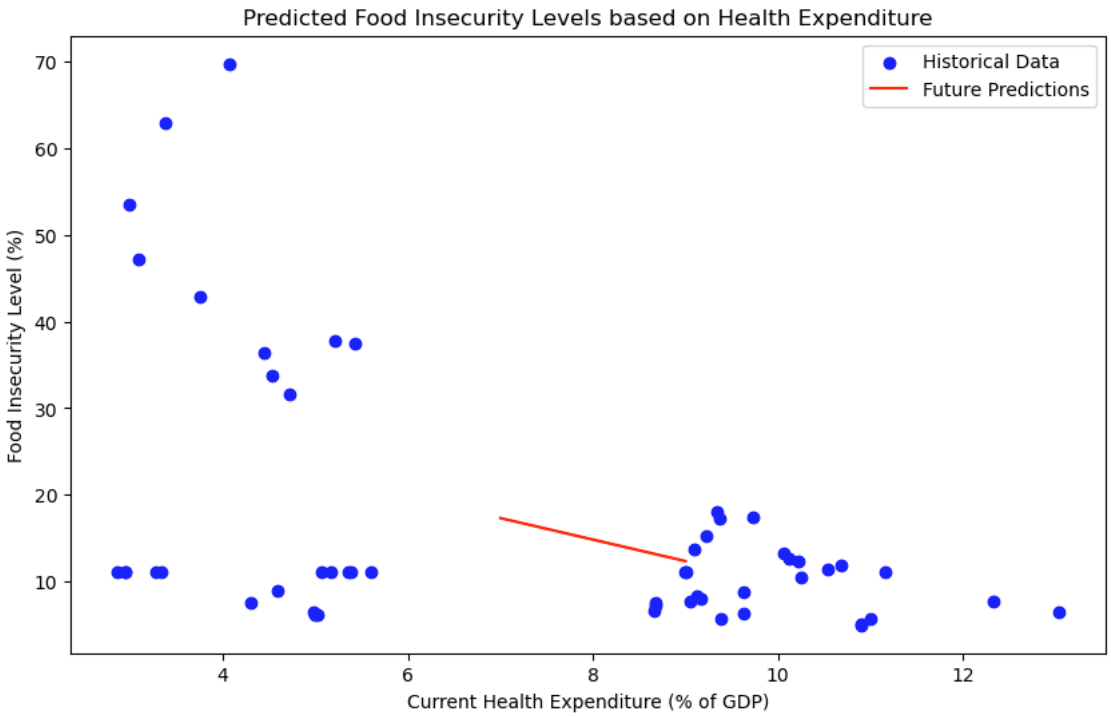

Figure 12: Predicted Food Insecurity Levels Based on Health Expenditure

	Current health expenditure	Predicted Food Insecurity Level (%)
0	7	17.335009
1	8	14.854103
2	9	12.373197

Table 3: Predicted Food Insecurity Levels Based on Current Health Expenditure

The forecast suggests a significant decline in food insecurity as health expenditure increases. This insight underscores the importance of continued investment in health

systems as a means to improve food security, particularly in countries with high food insecurity rates.

3.3 Forecasting Life Expectancy Based on Government Health Expenditure

The goal of this analysis is to predict future Life Expectancy based on trends in Government Health Expenditure (% of GDP). Understanding how government health investments affect life expectancy is crucial for strategic planning in public health.

A linear regression model was developed to predict Life Expectancy based on historical data for Domestic General Government Health Expenditure. Future predictions were made for the years 2024 to 2026, based on projected government health expenditure values.

The model predicts an increase in life expectancy as government health expenditure rises.

At 7% government health expenditure, life expectancy is forecasted to be 82.05 years.

By 9% expenditure, life expectancy is expected to reach 87.03 years.

The scatter plot below displays the historical data (blue) and the predicted life expectancy values (red) based on projected government health expenditure.

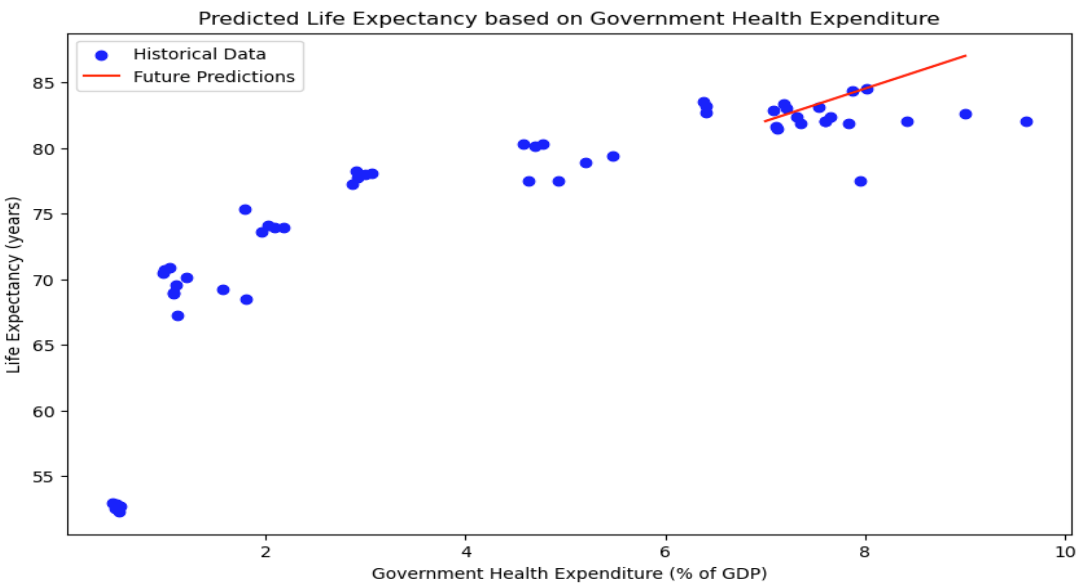

Figure 13: Predicted Life Expectancy Based on Government Health Expenditure for the Next 5 Years

```
   Domestic general government health expenditure
0                                               7
1                                               8
2                                               9

   Predicted Life Expectancy (years)
0                          82.052867
1                          84.543320
2                          87.033773
```

Table 4: Predicted Life Expectancy Based on Domestic Government Health Expenditure

The results indicate that increasing government health expenditure is strongly correlated with improvements in life expectancy. This highlights the critical role that sustained investment in public health systems plays in enhancing population health and longevity.

Conclusion

This report analysed health outcomes, food insecurity, food prices, and government health expenditures across ten selected countries, focusing on trends related to Sustainable Development Goal 3 (SDG 3): ensuring healthy lives and well-being for all. The analysis revealed several key insights.

1. Summary of Key Findings

- Government Health Expenditure: Countries with higher public health investment, such as Canada, Finland, and Australia, demonstrated better health outcomes, including higher life expectancy and lower maternal mortality rates. In contrast, lower-investment countries like Nigeria and Nepal faced greater health challenges.
- Food Insecurity: Significant disparities in food insecurity were found, with countries like Nigeria and Nepal experiencing high levels of insecurity, while countries such as Canada had much lower rates.
- Rising Food Prices: The cost of a healthy diet is expected to increase, especially in lower-income countries, potentially worsening food insecurity.

- Inverse Relationship: Higher government health expenditure is linked to lower food insecurity, highlighting the importance of public health investment in addressing food access.

2. Patterns Identified

1. Strong correlation between government health spending and improved health outcomes.
2. Countries with higher food insecurity, such as Nigeria, have lower health achievement and life expectancy.
3. Rising food prices may hinder health equity, especially in vulnerable nations.

3. Conclusion

Public health investment plays a crucial role in improving national health outcomes and reducing food insecurity. Countries with stronger public health systems experience better life expectancy and lower maternal mortality. To achieve SDG 3, particularly in lower-performing countries like Nigeria and Nepal, it is essential to increase health spending and address food insecurity. With sustained investment and targeted interventions, significant progress can be made toward improving global health outcomes.

Reference

- **United Nations, n.d.** Sustainable Development Goals. Available at: https://www.un.org/sustainabledevelopment/education/

Individual task and contribution

TASK	CONTRIBUTION
Data Preparation Phase	
Explore and define background and develop context	Phuong Linh Thai
Selection Data	Bang Vy Huynh
Data reparation	Phuong Linh Thai/ Bang Vy Huynh
Revision for data preparation	Bang Vy Huynh
EDA Phase	
Explore the dataset with python in Jupyter Notebook	Phuong Linh Thai/ Bang Vy Huynh
Detail explanation of the exploratory in the report	Phuong Linh Thai/ Bang Vy Huynh
Report	
Conclusion and final review report	Phuong Linh Thai/ Bang Vy Huynh
Presentation Slides	Phuong Linh Thai